Natural Language Processing with Deep Learning

Course Project Review

Moderation Mind

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Topic

Toxic Comment Moderation on Low-Resource Languages

This project develops a robust toxic comment moderation framework tailored for low-resource languages. Unlike traditional toxicity detection systems that rely heavily on large annotated corpora and standardized linguistic structures, the proposed system is designed to operate effectively under limited data availability, lexical variability, and informal user-generated content.

Uniqueness

Low-Resource-Centric Toxicity Modeling:

Most toxicity detection systems are developed for high-resource languages such as English. This project explicitly focuses on low-resource languages where labeled data is scarce, spelling variations are common, and linguistic tools are limited. The modeling decisions are tailored specifically for such constraints rather than adapting high-resource assumptions.

Robustness to Noisy Social Media Text

Low-resource online text often contains:

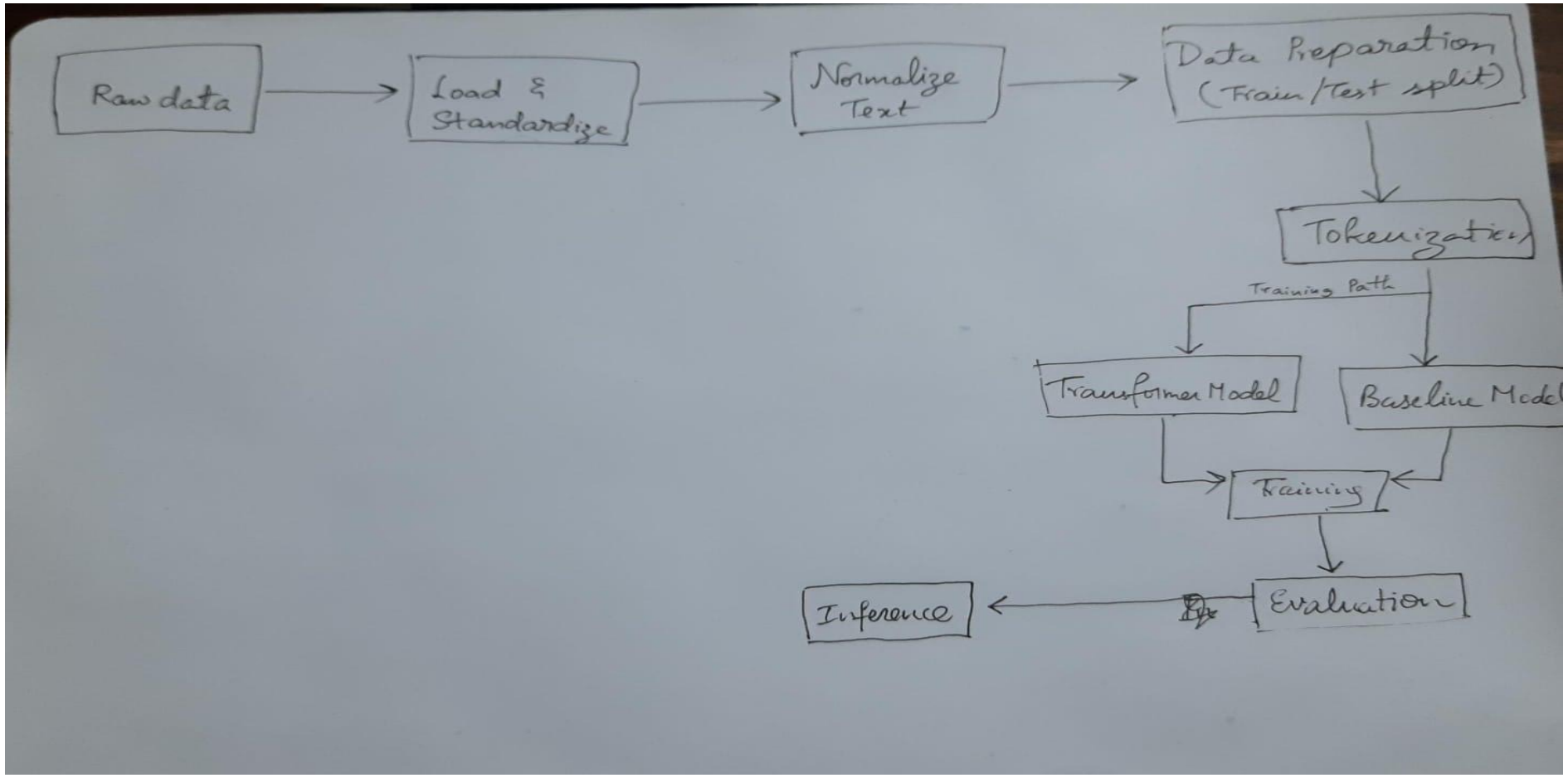
- spelling variations
- slang and informal grammar
- emojis and hashtags
- code-mixing
- character elongations

The proposed system includes preprocessing and normalization strategies specifically designed for handling such noise.

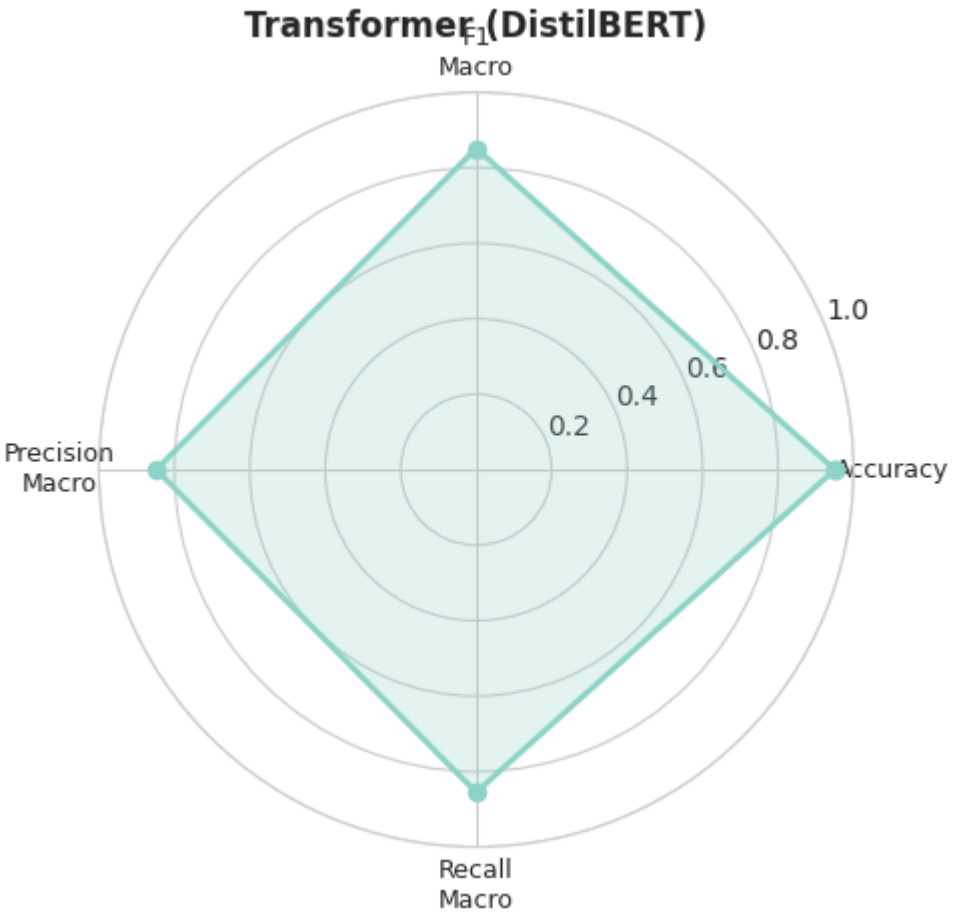
Dataset

Dataset	Link	Size	Attribute
Multilingual Toxic Comment Classification Dataset	https://github.com/AkshayaANANTAM/Multilingual-Toxic-comments-classification	5.03 GB	csv
Hate Speech and Offensive Language Datasets (Low-Resource Variants)	https://www.kaggle.com/datasets/mrmorj/hate-speech-and-offensive-language-dataset	2.55 MB	csv

Overall design or approach in a free hand diagram



Final results so far

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MODEL PERFORMANCE SUMMARY REPORT
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🏆 Best ACCURACY: Transformer (DistilBERT)
   Score: 0.9504

🏆 Best F1 MACRO: Transformer (DistilBERT)
   Score: 0.8516

🏆 Best PRECISION MACRO: Transformer (DistilBERT)
   Score: 0.8482

🏆 Best RECALL MACRO: Transformer (DistilBERT)
   Score: 0.8550

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OVERALL MODEL RANKING
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1. Transformer (DistilBERT)
   F1-Macro: 0.8516 | Accuracy: 0.9504 | Combined Score: 0.8911
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Done vs Remaining to be done ?

No	Description	Done or To be done
1	Data Preprocessing (Normalization, Tokenization)	Done
2	Fine-tuning DistilBERTa	Done
3	Training on Low-Resource Language	Done
4	Detecting spelling variation of toxic words	Partially Done
5	Detecting toxicity in long sentences	Done

Quantity and quality of work

no	Code functionality	% Complete	Runs without problem (Y/N)	If there are minor issues, indicate
1.	**Data Loading & Preprocessing Pipeline** - Loads multilingual toxic comment datasets, auto-detects column schemas, normalizes text (URL/mention replacement), handles romanized Hindi, applies leetspeak deobfuscation	100%	Yes	None
2.	**Label Parsing & Classification Setup** - Converts categorical labels to binary format, supports multilabel subtypes (abusive, hate_targeted, threat)	100%	Yes	None
3.	**Baseline Model (TF-IDF + Logistic Regression)** - Trains scikit-learn baseline with TF-IDF features (1-2 grams, 20K features) and balanced logistic regression	100%	Yes	None
4.	**Inference Module** - Model loading and prediction functionality for deploying trained models	90%	Yes	Minor: Interactive prediction features may need finalization

Top few learning

Serial No	Top learning in this project
1	I learnt how SLMs are fine-tuned and how fine-tuning actually works
2	I learnt how to handle Low-Resource Language including Normalization which contains less dataset available
3	I learnt how toxicity is predicted in different situations

Top unresolved challenges

Serial No	Brief description of unresolved challenges	Type of challenge (scope/data/design/implementation / others)
1	Model misclassifies implicit hate ('I hate <nationality>') as non-toxic	Dataset and Design issue
2	Toxic sentences without explicit abuse are classified as non-toxic	Design issue
3	Racial slurs apart from the "popular ones" are classified as non-toxic	Dataset issue
4	Sarcastic toxic sentences classified as non-toxic	Design issue

THANK YOU

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